



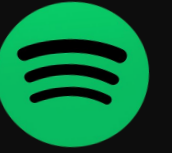
Spotify Popularity Prediction & Music Segmentation Analysis

Loïs

Selena

Tejpreet

Marketing Analytics – BUS6351



Business Understanding

Business Problem



-
- Spotify operates in a **highly competitive streaming environment**
 - Millions of tracks compete for limited visibility
 - Success depends on:
 - Playlist placement
 - Recommendation algorithms
 - User engagement

Core Problem:

- Difficult to determine which tracks will become successful
- Marketing and promotion decisions are often uncertain and inefficient

Business Objective

- Identify key drivers of song popularity
- Predict which tracks are likely to become hits
- Segment tracks for targeted marketing
- Support:
 - Promotion decisions
 - Playlist placement
 - Marketing strategy



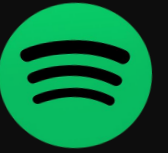
Analytical Approach

- Linear Regression → quantify popularity drivers
- Logistic Regression → classify hit vs non-hit tracks
- K-Means Clustering → identify track segments

Bottom line:

- Transform audio data into **actionable marketing insights**





Data Understanding & Preparation

Dataset Overview

Spotify Tracks Dataset

- Kaggle dataset: “Ultimate Spotify Tracks DB”
- Contains more than 232,000 Spotify tracks
- Includes audio features such as:
 - danceability
 - energy
 - loudness
 - acousticness
 - tempo
 - popularity
- Used for:
 - predictive modeling
 - classification analysis
 - clustering and segmentation

	genre	artist_name	track_name	track_id	popularity	acousticness	danceability	duration_ms	energy	instrumentalness	key	liveness	loudness	mode	speechiness	tempo	time_signature	valence
0	Movie	Henri Salvador	C'est beau de faire un Show	0BRjO6ga9RKCKjfDqeFgWV	0	0.611	0.389	99373	0.910	0.000	C#	0.3460	-1.828	Major	0.0525	166.969	4/4	0.814
1	Movie	Martin & les fées	Perdu d'avance (par Gad Elmaleh)	0BjC1NfoEOOusryehmNudP	1	0.246	0.590	137373	0.737	0.000	F#	0.1510	-5.559	Minor	0.0868	174.003	4/4	0.816
2	Movie	Joseph Williams	Don't Let Me Be Lonely Tonight	0CoSDzoNIKCRs124s9uTVy	3	0.952	0.663	170267	0.131	0.000	C	0.1030	-13.879	Minor	0.0362	99.488	5/4	0.368
3	Movie	Henri Salvador	Dis-moi Monsieur Gordon Cooper	0Gc6TVm52BwZD07Ki6tlvf	0	0.703	0.240	152427	0.326	0.000	C#	0.0985	-12.178	Major	0.0395	171.758	4/4	0.227
4	Movie	Fabien Nataf	Ouverture	0lusiXpMROHdEPvSI1fTQK	4	0.950	0.331	82625	0.225	0.123	F	0.2020	-21.150	Major	0.0456	140.576	4/4	0.390

Data Understanding

-
- Understanding the Dataset
 - Original dataset:
 - 232,725 observations
 - 18 variables
 - Combination of:
 - categorical variables
 - numerical variables
 - Only 1 missing value identified
 - More than 55,000 duplicate records discovered

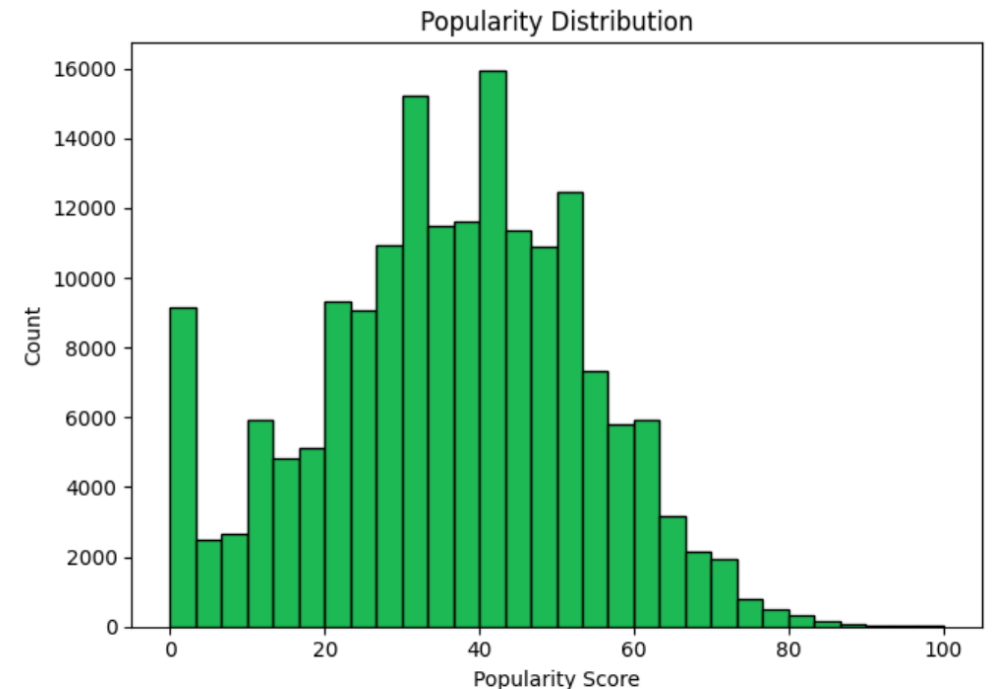
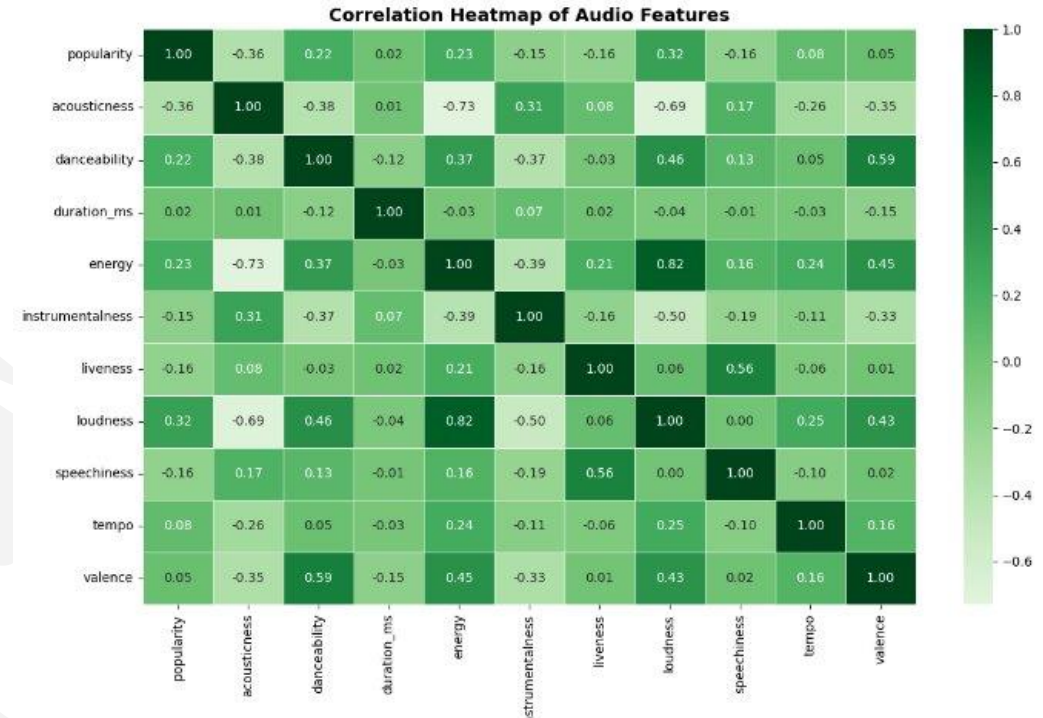
```
# Drop the 1 missing track_name row
df = df.dropna(subset=['track_name'])

# Drop duplicate tracks (same track_id)
df = df.drop_duplicates(subset=['track_id'])
```

```
Shape after cleaning: (176773, 18)
```


Exploratory Analysis Findings

- Exploratory Data Analysis
- Popularity distribution was highly uneven
- Most tracks had low to moderate popularity
- Energy and loudness showed strong positive relationships
- Acousticness showed negative relationships with energy and loudness
- Popularity influenced by combinations of audio features



Data Preparation

- Data Cleaning & Preprocessing
 - Removed incomplete observations using `dropna()`
 - Removed duplicate tracks using `drop_duplicates()`
 - Encoded categorical variables using `LabelEncoder`
 - Standardized numerical variables using `StandardScaler`
 - Prepared dataset for machine learning analysis
-

```
# Drop the 1 missing track_name row
df = df.dropna(subset=['track_name'])

# Drop duplicate tracks (same track_id)
df = df.drop_duplicates(subset=['track_id'])

# Check basic stats
print("Shape after cleaning:", df.shape)
print("\nPopularity Stats:")
print(df['popularity'].describe())
```

Shape after cleaning: (176773, 18)

Popularity Stats:

count	176773.000000
mean	36.273119
std	17.391056
min	0.000000
25%	25.000000
50%	37.000000
75%	49.000000
max	100.000000

Name: popularity, dtype: float64

Prepared Dataset Summary

Final Prepared Dataset

Cleaned and structured analytical dataset

Reduced inconsistencies and duplicate bias

Improved compatibility with:

- regression models
- classification models
- clustering analysis

Supported predictive modeling and audience segmentation

Modeling & Results

Spotify Marketing Analytics

- Multiple Linear Regression
- Logistic Regression
- K-Means Clustering (STG)
- Results & Evaluation



Three Models, One Goal

What drives a song's popularity on Spotify?



Linear Regression

Predict the exact popularity score (0–100)
from audio features

Analysis 1



Logistic Regression

Classify tracks as Hit or Non-Hit — a go/no-go
decision for release

Analysis 2



K-Means Clustering

Segment tracks into audience profiles for
targeted marketing (STG)

Analysis 3



Predicting the Popularity Score

Multiple linear regression on continuous numeric target (0–100)

Features Used as Predictors

- danceability** — how suitable for dancing (0–1)
- energy** — intensity and activity level (0–1)
- loudness** — overall loudness in dB
- acousticness** — confidence track is acoustic (0–1)
- valence** — musical positiveness (0–1)
- tempo** — beats per minute (BPM)
- speechiness** — presence of spoken words (0–1)
- instrumentalness** — likelihood of no vocals (0–1)
- liveness** — presence of live audience (0–1)
- key, mode, time_sig** — musical structure (encoded)

Methodology

- 80% training / 20% testing split
- Label encoding for categorical features
- Coefficients show impact direction & size

Evaluation Metrics

R²

% of popularity variance explained

RMSE

Avg prediction error in popularity pts

MAE

Mean absolute prediction error



What the Model Found

Feature coefficients and model performance

0.20

R^2 Score

15.57

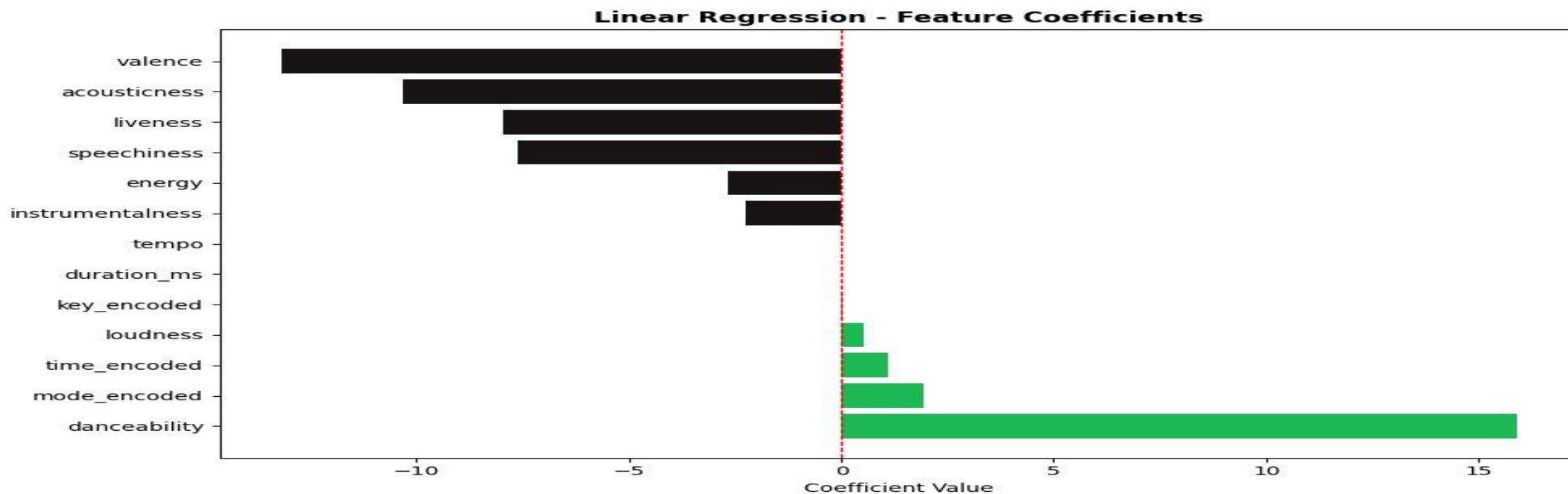
RMSE

~12.5

MAE



Audio features explain 20% of popularity. Artist fame, marketing & playlist placement drive the rest.





Classifying Hit vs. Non-Hit Tracks

Binary classification — will this track succeed commercially?

Defining a Hit

Hit = 1
popularity ≥ 50

Non-Hit = 0
popularity < 50

→ 23% Hits | 77% Non-Hits

⚠ Why threshold = 50, not 70?

- At threshold 70: only 2.2% of tracks were hits
- Model predicted everything as Non-Hit
- Got 97.8% accuracy — but F1 = 0, useless!
- Lowering to 50 gives balanced 77/23 split
- `class_weight='balanced'` applied to handle remaining imbalance

How It Differs from Linear Regression

	Linear	Logistic
Output	Score 0–100	Hit / Non-Hit
Question	How popular?	Will it succeed?
Use Case	Benchmarking	Release filter
Our Result	$R^2=0.20$	Recall=75%



Model Performance & Key Findings

Recall of 75% — catches 3 out of every 4 real hits

62.0%

Accuracy

35.1%

Precision

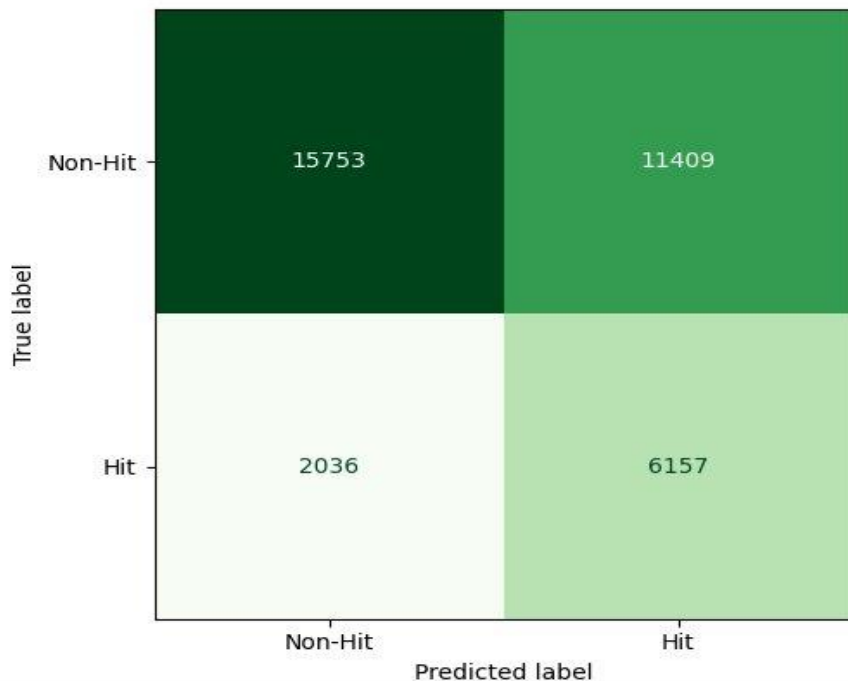
75.2%

Recall ★

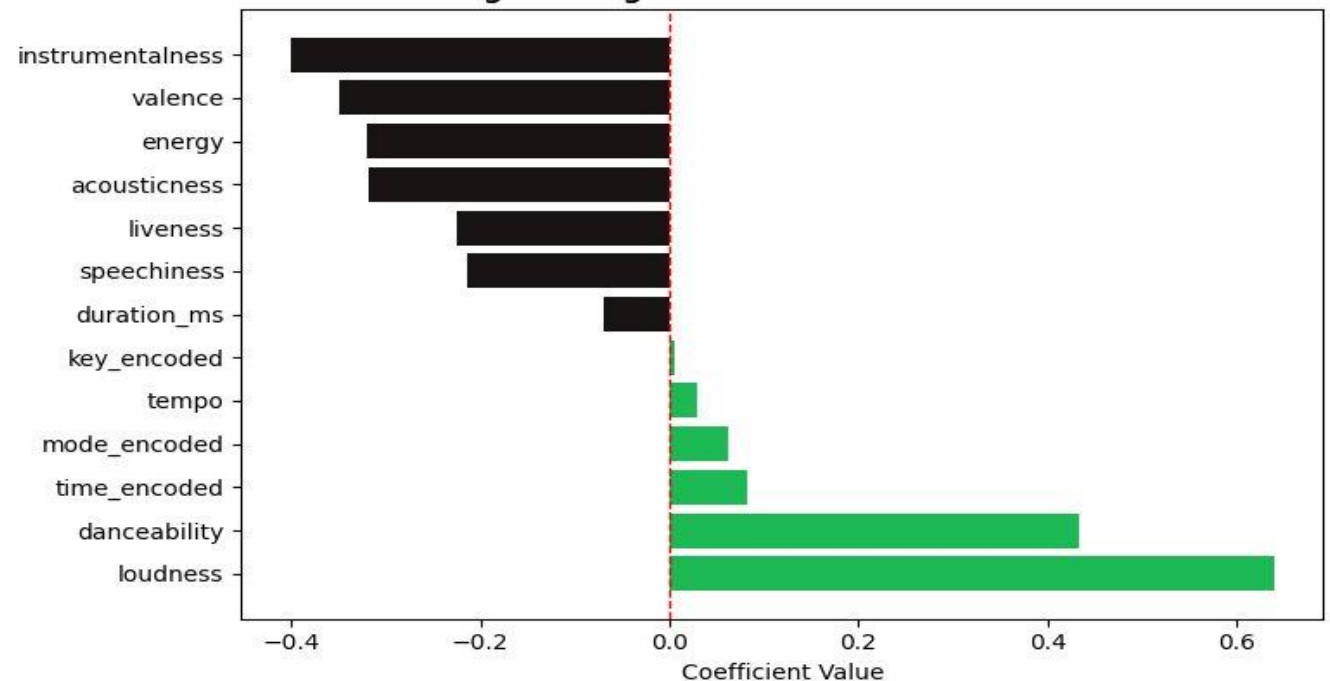
0.478

F1-Score

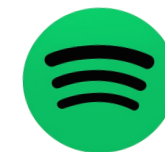
Confusion Matrix



Logistic Regression - Feature Coefficients



High recall (75%) is most valuable — catching real hits matters more than avoiding false alarms for release decisions.



Segmenting Tracks by Audio Profile

Unsupervised learning — discovering natural listener groups

S Segmentation

T Targeting

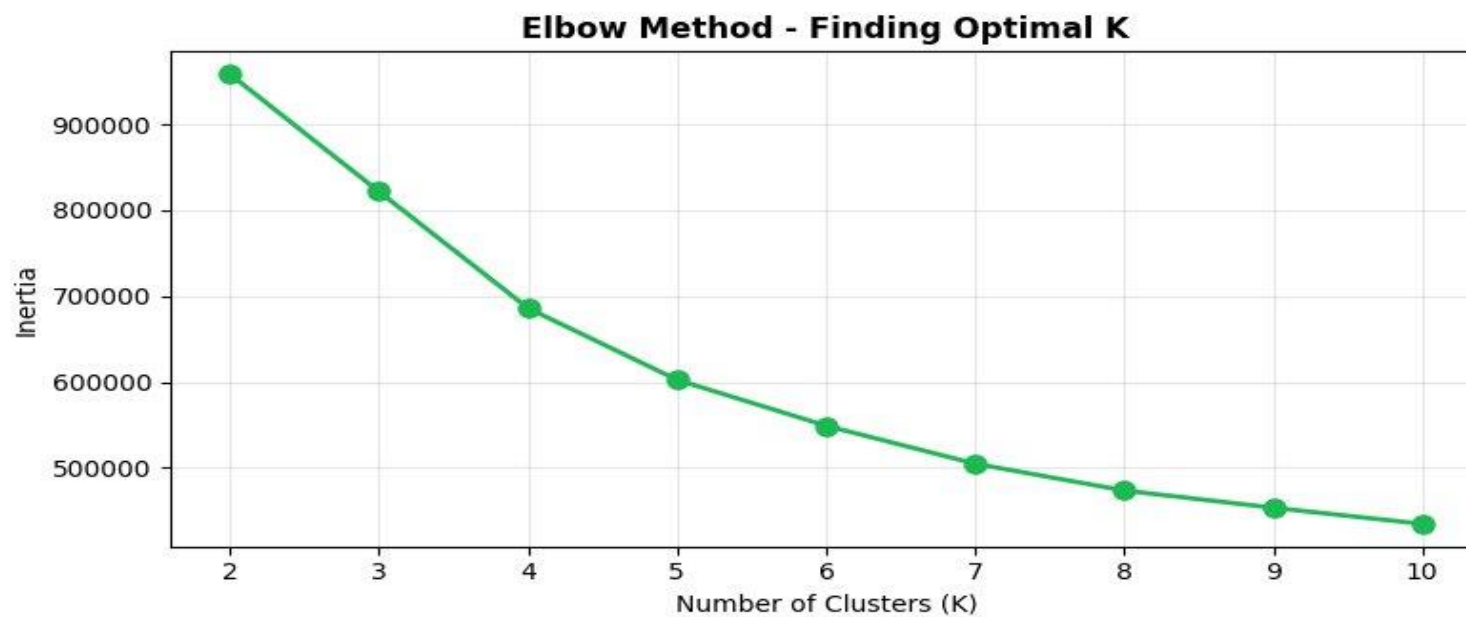
G Go-to-Market Positioning

Identify natural segments → Find highest-value targets → Design tailored campaigns

Clustering Features

- danceability
- energy
- acousticness
- instrumentalness
- valence
- tempo
- loudness
- speechiness

Elbow Method — Finding Optimal K





5 Distinct Audience Segments Identified

Each cluster represents a unique sonic identity and marketing opportunity

Feel-Good Dance

High danceability (0.70) · High valence (0.66) · Loud

40.7

PRIMARY TARGET

High Energy Electronic

Highest energy (0.76) · Fast tempo (143 BPM) · Low acousticness

40.5

PRIMARY TARGET

Soft Acoustic Pop

High acousticness (0.76) · Low energy (0.30)

32.2

Secondary

Ambient Instrumental

High instrumentalness (0.78) · High acousticness (0.84)

28.3

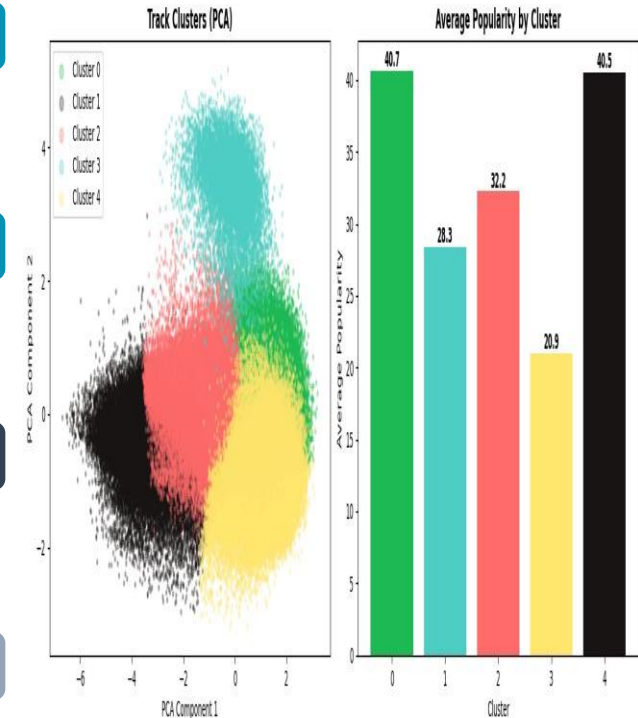
Niche

Spoken Word / Live

Very high speechiness (0.89) · Live recordings

20.9

Low Priority





From Clusters to Marketing Strategy

Segmentation → Targeting → Positioning

S

Segmentation

5 distinct sonic audience segments identified from 176,773 tracks — Feel-Good Dance, High Energy Electronic, Soft Acoustic Pop, Ambient Instrumental, and Spoken Word/Live.

T

Targeting

Clusters 0 and 4 are the highest-value targets — both score above 40 in average popularity and align perfectly with the top predictors from regression: high danceability, loudness, and energy.

G

Go-to-Market Positioning

Cluster 0 (Dance): social playlists, party contexts, brand partnerships. Cluster 4 (Electronic): workout, gaming, festival. Clusters 1 & 3: niche/independent channels only.



What We Found Across All Three Models

Consistent signals point to the same audio drivers of popularity

Model	Key Metrics	Best Used For	Top Finding
Linear Regression	$R^2=0.20$ RMSE=15.57	Benchmarking track potential	danceability = strongest driver
Logistic Regression	Recall=75% F1=0.48	Hit / non-hit release filter	loudness = top hit predictor
K-Means Clustering	K=5 meaningful segments	Audience targeting (STG)	Dance & Electronic = top targets



Overall Finding

All three models independently identify the same core traits: high danceability, high loudness, and high energy. When three different methods agree, that's strong convergent evidence these audio features genuinely drive commercial success.

Limitations

- $R^2=0.20$ — 80% unexplained (artist fame, marketing)
- Popularity score decays over time — recency bias
- Genre not included as a regression feature

Coming Next

- Ridge/Lasso Regression — handle multicollinearity
- Random Forest — improve hit classification
- Add external data: charts, artist fame, social media



What External Data Would Improve Our Models?

Closing the gap on the 80% of popularity our audio features can't explain

Spotify Charts API

Adds: Weekly top 200 rankings by country

Impact: Stronger hit definition beyond the 0–100 score

Release Date Data

Adds: Track age, release timing

Impact: Accounts for recency bias in Spotify's scoring

Billboard Hot 100

Adds: US chart rankings across genres

Impact: Industry benchmark to validate our hit threshold

TikTok / Instagram

Adds: Viral mentions, hashtag counts

Impact: Captures viral marketing — the missing 80% of R^2

MusicBrainz / Last.fm

Adds: Artist play counts & listener counts

Impact: Separates audio quality from artist brand effect

IFPI / Luminate Reports

Adds: Genre market share & revenue

Impact: Weights genre importance in clustering models

Key Takeaways

- 01 Danceability, loudness & energy**
are the top popularity drivers — consistent across all 3 models
- 02 Linear Regression ($R^2=0.20$)**
audio features explain ~20%; artist fame & marketing drive the rest
- 03 Logistic Regression (Recall=75%)**
catches 3 out of 4 real hits — a reliable release filter
- 04 K-Means: 5 segments found**
Clusters 0 & 4 (Dance + Electronic) are highest-value marketing targets

What We Will Do Differently

Planned improvements as the course progresses and new techniques are introduced

01 Ridge & Lasso Regression

Handle multicollinearity between energy and loudness ($r=0.82$). Lasso will auto-select the most relevant features and reduce noise in the model.

02 Random Forest Classifier

Replace logistic regression with a tree-based model to capture non-linear relationships — expected to push F1-Score well beyond 0.48.

03 Genre-Stratified Models

Run separate models per genre. A pop track and a classical track behave very differently — one model for all 26 genres is too broad.

04 Add Artist-Level Data

Incorporate Last.fm artist play counts to separate audio quality from artist fame — directly targeting the missing 80% of R^2 .

05 Social & Viral Signals

Include TikTok and Instagram mention counts per track to capture viral marketing effects that audio features alone cannot explain.

06 Dynamic Popularity Score

Adjust for recency bias — Spotify scores decay over time. Normalizing by track age will make comparisons across release years fairer.

How AI Can Take This Further

Six ways artificial intelligence can extend and strengthen this project beyond traditional statistical models

01

Generative AI for Feature Engineering

Use LLMs to extract features from lyrics and social media — sentiment, theme, emotional tone — to improve R^2 beyond 0.20.

02

NLP for Sentiment Analysis

Apply BERT or GPT models to listener reviews and social posts to turn qualitative buzz into numeric popularity predictors.

03

Deep Learning for Popularity Prediction

Neural networks can capture complex non-linear feature interactions — potentially pushing recall well beyond the current 75%.

04

Reinforcement Learning for Campaign Optimization

An RL agent could learn optimal budget allocation across our 5 audience segments — turning static clusters into a dynamic marketing engine.

05

AI-Powered A&R Recommendation

Collaborative filtering models could predict which specific listener segments a new track will resonate with — far more granular than hit/non-hit.

06

AutoML for Model Optimization

AutoML frameworks like Google AutoML or H2O.ai could search hundreds of model configurations automatically across 176K tracks.



Recommendations

Content Strategy



Prioritize tracks with:

High danceability
High energy
High loudness



Reduce promotion of:

Acoustic / instrumental tracks



Apply as pre-release screening criteria

Use Hit Prediction to Drive Promotion



Implement model as a **release decision tool**



Automatically flag high-potential tracks



Allocate marketing budget to predicted hits



Accept over-promotion to avoid missing hits

Focus Marketing on High-Value Segments

Prioritize

Prioritize marketing for:

- Feel-Good Dance
- High Energy Electronic

Reduce

Reduce investment in low-performing segments

Align

Align content investment with top clusters

Execute Context-Based Marketing

Match	Use	Promote
Match campaigns to listening context: • Social / viral (Cluster 0) • Workout / gaming (Cluster 4)	Use playlists as primary marketing channel	Promote tracks early in lifecycle



Expand Data to Improve Decision-Making

- Integrate:
 - Artist popularity data
 - Social media signals
 - Release timing
 - Models into Spotify workflow
- Continuously retrain models
- Use A/B testing to validate decisions
- Use external data to improve prediction accuracy



Conclusion



Project Conclusion

Spotify Marketing Analytics — End-to-End Summary

1

Business Problem & Data

176,773 Spotify tracks across 26 genres with 18 audio features. The core question: can audio characteristics predict commercial success? Data was sourced from Kaggle and cleaned by removing 1 null and ~56,000 duplicate tracks.

2

Exploratory Data Analysis

EDA revealed Pop, Rap and Rock as the most popular genres. Loudness ($r=0.32$) and danceability ($r=0.22$) showed the strongest correlations with popularity. Energy and loudness were highly correlated ($r=0.82$) — a multicollinearity flag.

3

Modeling — Three Analyses Built

Linear regression ($R^2=0.20$, $RMSE=15.57$) predicted popularity scores. Logistic regression achieved 75% recall classifying hits. K-Means clustering identified 5 distinct audience segments with $K=5$ confirmed by the elbow method.

4

Key Finding Across All Models

Danceability, loudness, and energy are the top popularity drivers — consistent across all three models independently. Audio features alone explain ~20% of popularity; artist fame and marketing account for the remaining 80%.

5

Recommendations

Target Feel-Good Dance and High Energy Electronic segments (avg popularity 40+) for marketing campaigns. Prioritize loud, danceable, high-energy tracks for release. Future work: add artist data, social signals, and Ridge/Random Forest models.

176K

Tracks
Analysed

3

Models
Built

75%

Hit
Recall

5

Audience
Segments



Thank You

Loïs

Selena

Tejpreet

Marketing Analytics – BUS6351

References

- Aguiar, L., & Waldfogel, J. (2018). Platforms, promotion, and product discovery: Evidence from Spotify playlists. *National Bureau of Economic Research*.
- International Federation of the Phonographic Industry. (2025). *Global Music Report 2025*.
- Spotify. (2025). *Spotify Annual Report 2025*.
- Spotify for Artists. (2025). *How playlists and recommendations support music discovery*.
- Kaggle. (2026). *Ultimate Spotify Tracks DB dataset*. <https://www.kaggle.com/datasets/zaheenhamidani/ultimate-spotify-tracks-db>